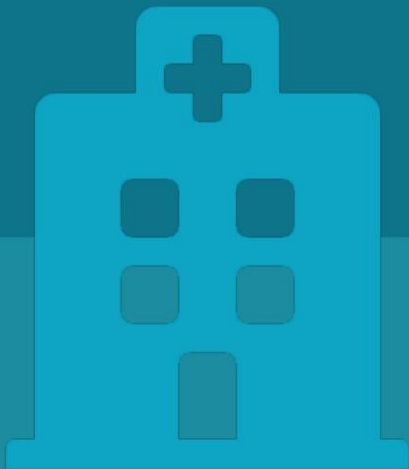




MediCare Pro

Smart Healthcare Management System

DEPI Graduation Project · 2026

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Full-Stack Web Application

Agenda

Presentation outline

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Project Overview

What is MediCare Pro?

MediCare Pro is a full-stack healthcare management platform that digitizes appointment booking for patients, doctors, and administrators — replacing manual phone-based scheduling with a modern, secure web application.

3

User Roles

500+

Doctors

50K+

Patients

24/7

Support



Doctor Search & Filtering



Online Appointment Booking



Role-Based Authorization



JWT Secure Authentication



Admin Analytics Dashboard



Weekly Schedule Management

Problem Statement

Why MediCare Pro was built



Appointment Conflicts

Phone-based scheduling leads to double-bookings and time conflicts across clinics



Poor Communication

No direct digital channel between patients and doctors for updates or follow-ups



No Discovery

Patients struggle to find verified specialists filtered by specialty and rating



No Centralization

Scattered records make platform-wide monitoring and analytics impossible

✦ MediCare Pro solves these problems with a secure, centralized, modern web platform

Project Objectives

What MediCare Pro aims to achieve

1

Digitize Healthcare

Replace manual phone bookings with a modern online appointment platform

2

Improve Communication

Enable direct doctor-patient interaction and appointment tracking

3

Reduce Conflicts

Smart scheduling prevents double-bookings and time clashes automatically

4

Centralize Management

One platform for patients, doctors, and admins to operate from seamlessly

5

Responsive UX

Intuitive interface accessible from any device with a consistent experience

Target Users

Three distinct roles with purpose-built interfaces



- Register & Login
- Search & Filter Doctors
- Book Appointments
- Cancel Appointments
- Leave Reviews
- Manage Profile



- Register for Approval
- Manage Weekly Schedule
- Accept / Reject Appointments
- Complete Appointments
- Manage Professional Profile



- Manage All Users
- Approve Doctor Registrations
- Analytics Dashboard
- Monitor All Appointments
- System-wide Management

Main Features

Core functionalities of the platform



Authentication

JWT tokens, bcrypt passwords, protected routes, session persistence



Doctor Discovery

Search by name or specialty, filter by rating, sort by availability



Appointment Booking

Select date & slot, add notes, instant booking confirmation



Doctor Management

Weekly schedule editor with per-day slot durations, approval workflow



Admin Analytics

Revenue tracking, appointment breakdown charts, user statistics



Reviews System

Star ratings and patient comments tied to completed visits

Technology Stack

Modern, industry-standard technologies

Frontend

React.js 18

Vite (Build Tool)

JavaScript ES6+

React Router DOM

Context API

Axios

CSS3

Backend

Node.js

Express.js

SQL Server

JWT Authentication

bcrypt

REST APIs

CORS & Middleware

System Architecture

How the layers connect



React Frontend

Components · Context API · React Router DOM



Axios HTTP Layer

REST Calls · Interceptors · Authorization Headers



Express REST API

Controllers · Middleware · Route Guards



SQL Server Database

Tables · Stored Procedures · Views

Database Overview

SQL Server schema — key entities & relationships



Users

🔑 UserID (PK)

Name

Email

PasswordHash

Role

Phone



Doctors

🔑 DoctorID (PK)

UserID (FK)

Specialty

Fee

LicenseNumber

Status



Appointments

🔑 AppointmentID (PK)

PatientID (FK)

DoctorID (FK)

DateTime

Status



Availability

🔑 AvailabilityID (PK)

DoctorID (FK)

DayOfWeek

StartTime

SlotDuration

Security Features

Enterprise-grade security from day one



JWT Authentication

Stateless tokens with expiry for secure session management



Password Hashing

bcrypt with salt rounds — passwords never stored in plain text



Role-Based Access

Patient, Doctor, Admin — each with isolated routes and permissions



Protected Routes

Frontend route guards redirect unauthenticated users automatically



Doctor Verification

New doctors require admin approval before accessing the platform

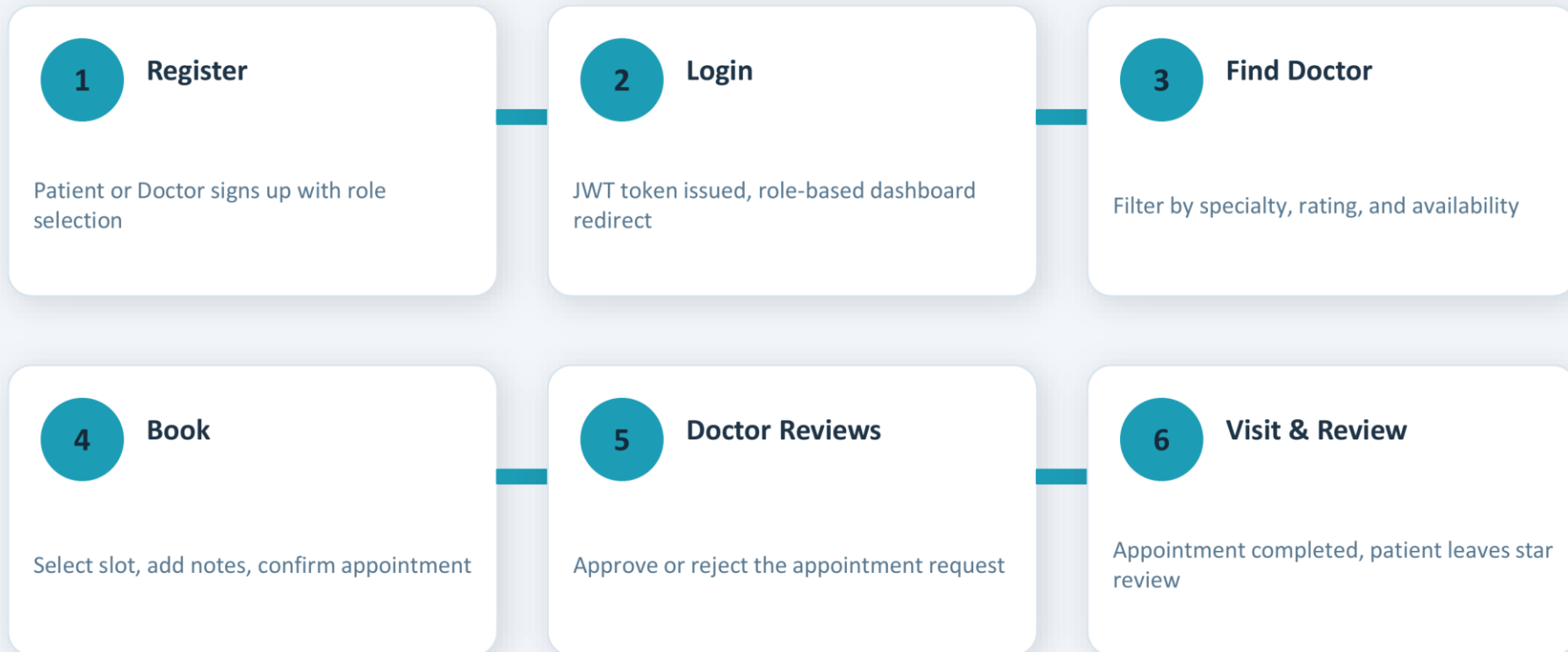


Secure API Requests

Authorization headers on all protected endpoints via Axios interceptors

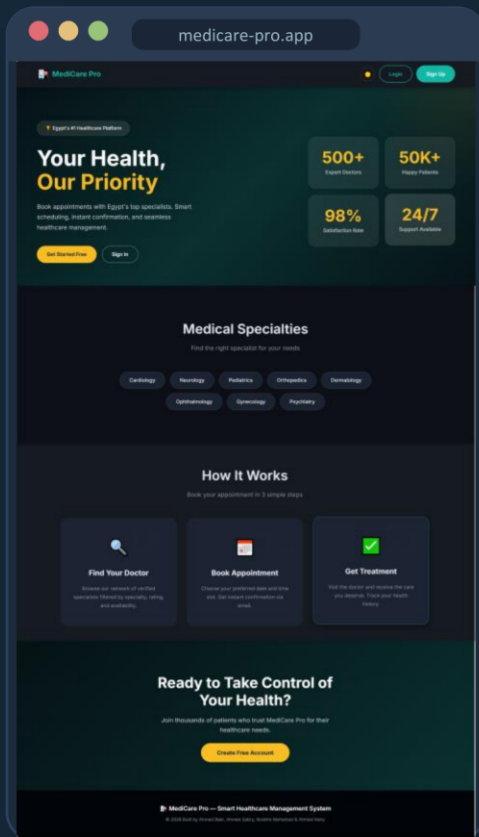
Application Workflow

End-to-end user journey in 6 steps



UI Showcase — Landing Page

Egypt's #1 Healthcare Platform



01 Hero Section

Tagline, stats (500+ Doctors, 50K+ Patients), CTA buttons

02 Medical Specialties

Browse 8 specialty categories with pill-style navigation

03 How It Works

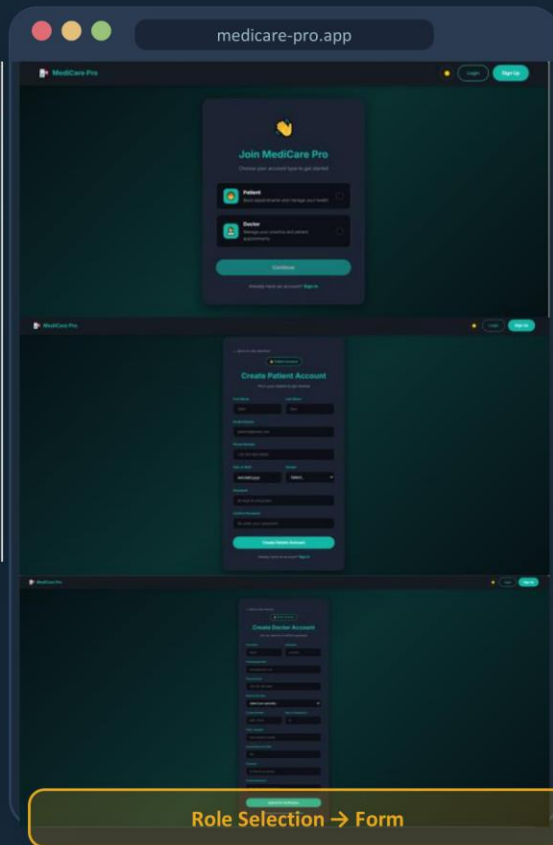
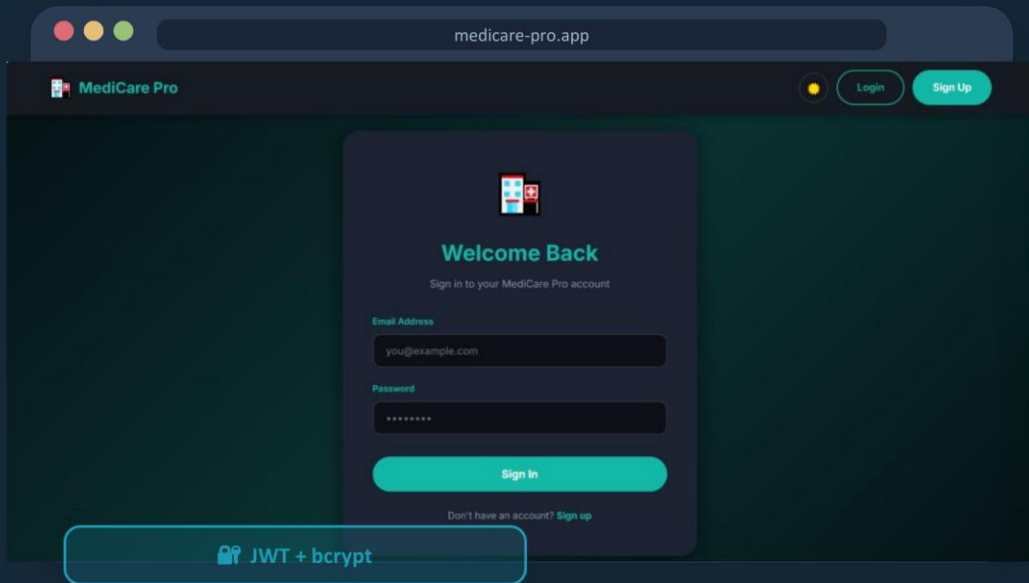
3-step visual guide: Find → Book → Get Treatment

04 Call to Action + Footer

Registration prompt and team footer

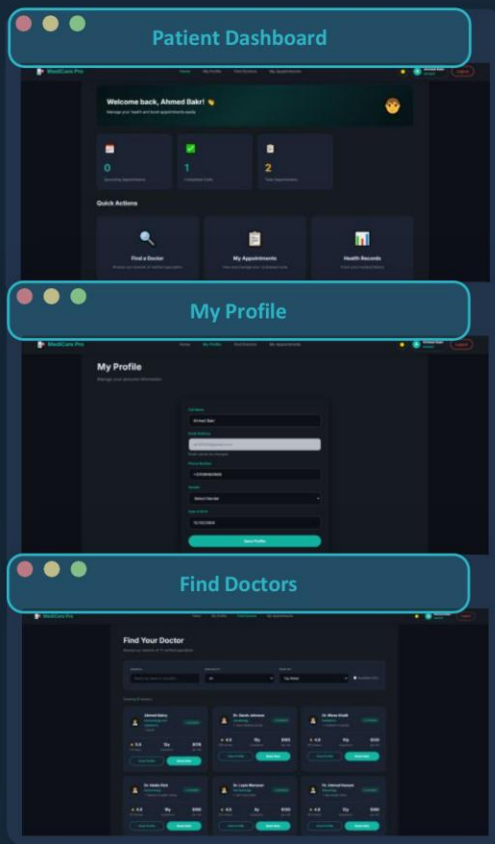
UI Showcase — Authentication

Secure login and multi-step registration



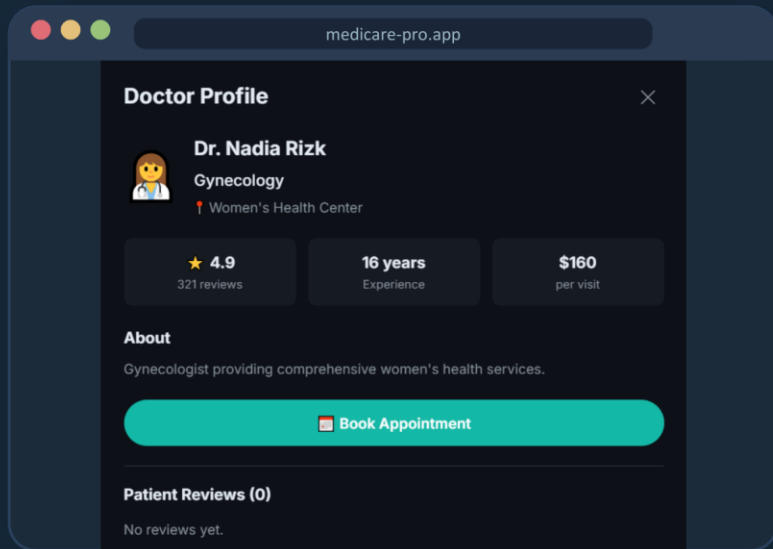
UI Showcase — Patient Module

Dashboard, Profile & Doctor Discovery

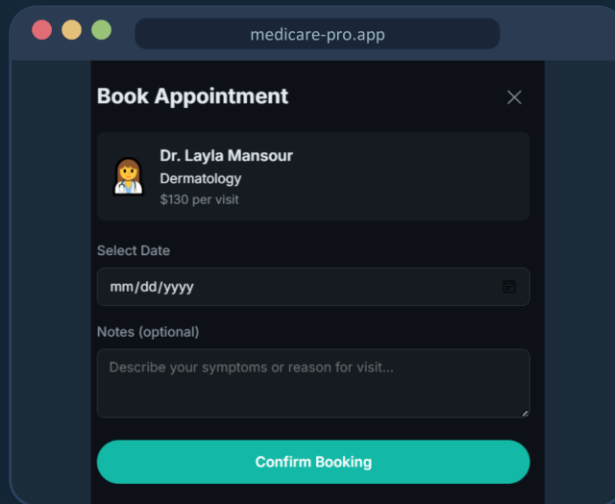


UI Showcase — Patient Module (cont.)

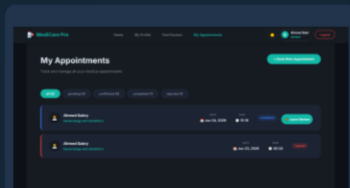
Booking flow and appointment history



Doctor Profile Popup



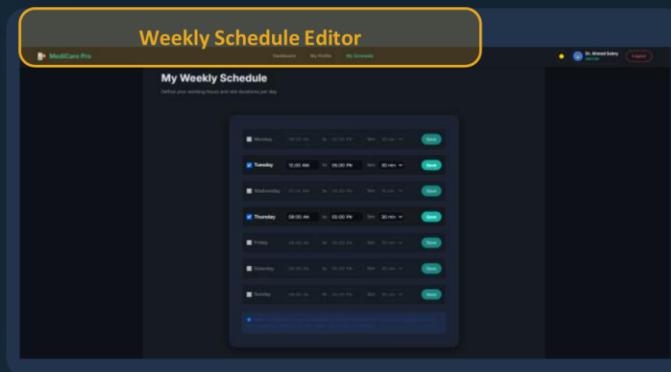
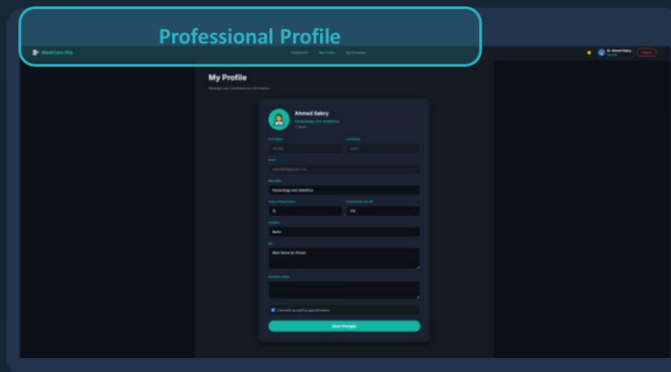
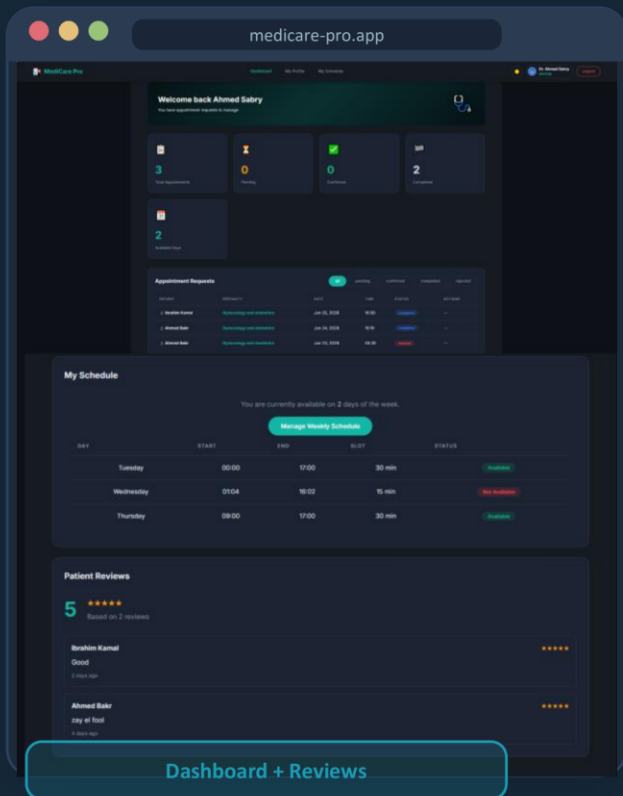
Book Appointment Modal



My Appointments — Status tracking

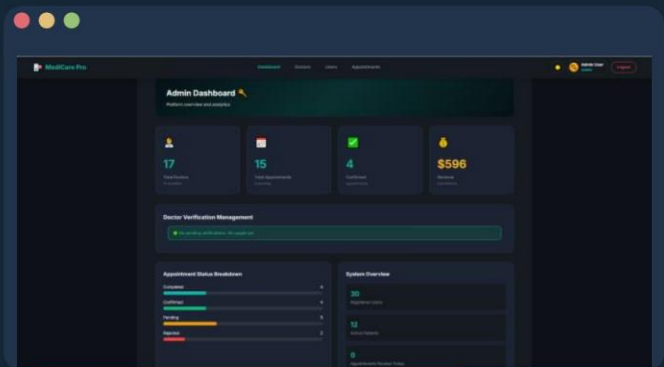
UI Showcase — Doctor Module

Dashboard, profile & schedule management



UI Showcase — Admin Dashboard

Complete platform oversight and management



Admin Dashboard

The Manage Doctors page allows for viewing and managing all registered doctors. It includes a search bar to filter by name, email, or specialty. The table below lists the doctors with their details.

DOCTOR NAME	EMAIL	SPECIALTY	EXPERIENCE	AGE	STATUS	APPOINTMENTS	ACTIONS
1. Dr. Sarah Johnson	sarah@medpro.com	Cardiology	10y	35	Active	12	View
2. Dr. Ahmed Hassan	ahmed@medpro.com	Neurology	15y	40	Active	8	View
3. Dr. Maria Khadi	maria@medpro.com	Pediatrics	12y	30	Active	15	View
4. Dr. Omar Farouk	omar@medpro.com	Orthopedics	8y	28	On Hold	5	View
5. Dr. Layla Mansour	layla@medpro.com	Dermatology	6y	25	Active	3	View
6. Dr. Karim Nabil	karim@medpro.com	Ophthalmology	18y	45	Active	10	View
7. Dr. Huda Rida	huda@medpro.com	Gynecology	10y	32	Active	7	View
8. Dr. Youssef Tamer	youssef@medpro.com	Psychiatry	14y	38	On Hold	2	View

Manage Doctors

The Manage Users page allows for viewing and managing all platform users. It includes a search bar to filter by name or email. The table below lists the users with their details.

USER NAME	EMAIL	ROLE	STATUS	APPOINTMENTS	ACTIONS
Admin User	admin@medpro.com	Admin	Active	0	View
1. Dr. Sarah Johnson	sarah@medpro.com	Doctor	Active	12	View
2. Dr. Ahmed Hassan	ahmed@medpro.com	Doctor	Active	8	View
3. Dr. Maria Khadi	maria@medpro.com	Doctor	Active	15	View
4. Dr. Omar Farouk	omar@medpro.com	Doctor	On Hold	5	View
5. Dr. Layla Mansour	layla@medpro.com	Doctor	Active	3	View
6. Dr. Karim Nabil	karim@medpro.com	Doctor	Active	10	View
7. Dr. Huda Rida	huda@medpro.com	Doctor	Active	7	View
8. Dr. Youssef Tamer	youssef@medpro.com	Doctor	On Hold	2	View

Manage Users

The All Appointments page allows for monitoring and managing all appointments across the platform. It includes a search bar to filter by patient, doctor, or specialty. The table below lists the appointments with their details.

PATIENT	DOCTOR	SPECIALTY	DATE	TIME	FEE	STATUS	ACTIONS
1. Ibrahim Kamal	Hasanah Mohamed	Cardiology	6/26/2028	18:30	\$0	Confirmed	Cancel
2. Ibrahim Kamal	Hasanah Mohamed	Cardiology	6/26/2028	12:30	\$0	Pending	Cancel
3. Ibrahim Kamal	Ahmed Sabry	Cardiology	6/25/2028	10:30	\$0	Completed	Cancel
4. Ahmed Baki	Ahmed Sabry	Cardiology	6/24/2028	15:15	\$0	Completed	Cancel

All Appointments

Project Strengths

What sets MediCare Pro apart



Secure by Design

JWT + bcrypt + RBAC implemented from day one — not bolted on



Multi-Role Platform

Separate, purpose-built dashboards for each user type



Analytics Ready

Admin dashboard with lifetime revenue tracking and charts



Smart Scheduling

Weekly availability slots with custom durations per day



Reviews System

Star ratings tied to completed visits for quality control



Modern Tech Stack

React + Vite + Node.js + SQL Server — industry standard

Future Enhancements

Planned features for the next development phase



Online Payments

Integrated payment gateway for consultation fees



Video Consultation

Real-time teleconsultation between patient and doctor



Notifications

Email & SMS alerts for appointment reminders



AI Doctor Matching

Smart doctor matching based on symptoms & history



Medical Records

Digital health records and prescription management



Mobile Application

Native iOS & Android apps for on-the-go access

Conclusion

What we accomplished

MediCare Pro successfully delivers a modern, full-stack healthcare management system that addresses real-world problems in appointment booking and clinic management.

✓ Complete role-based platform for Patients, Doctors, and Administrators

✓ Secure JWT authentication with bcrypt password hashing

✓ Responsive React.js frontend with elegant dark-mode UI

✓ RESTful Node.js + Express backend connected to SQL Server

✓ Scalable architecture ready for future enhancements



Thank You!

MediCare Pro — Smart Healthcare Management System



GitHub Repository

github.com/A200B5/smart-healthcare-system



DEPI Graduation Project

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